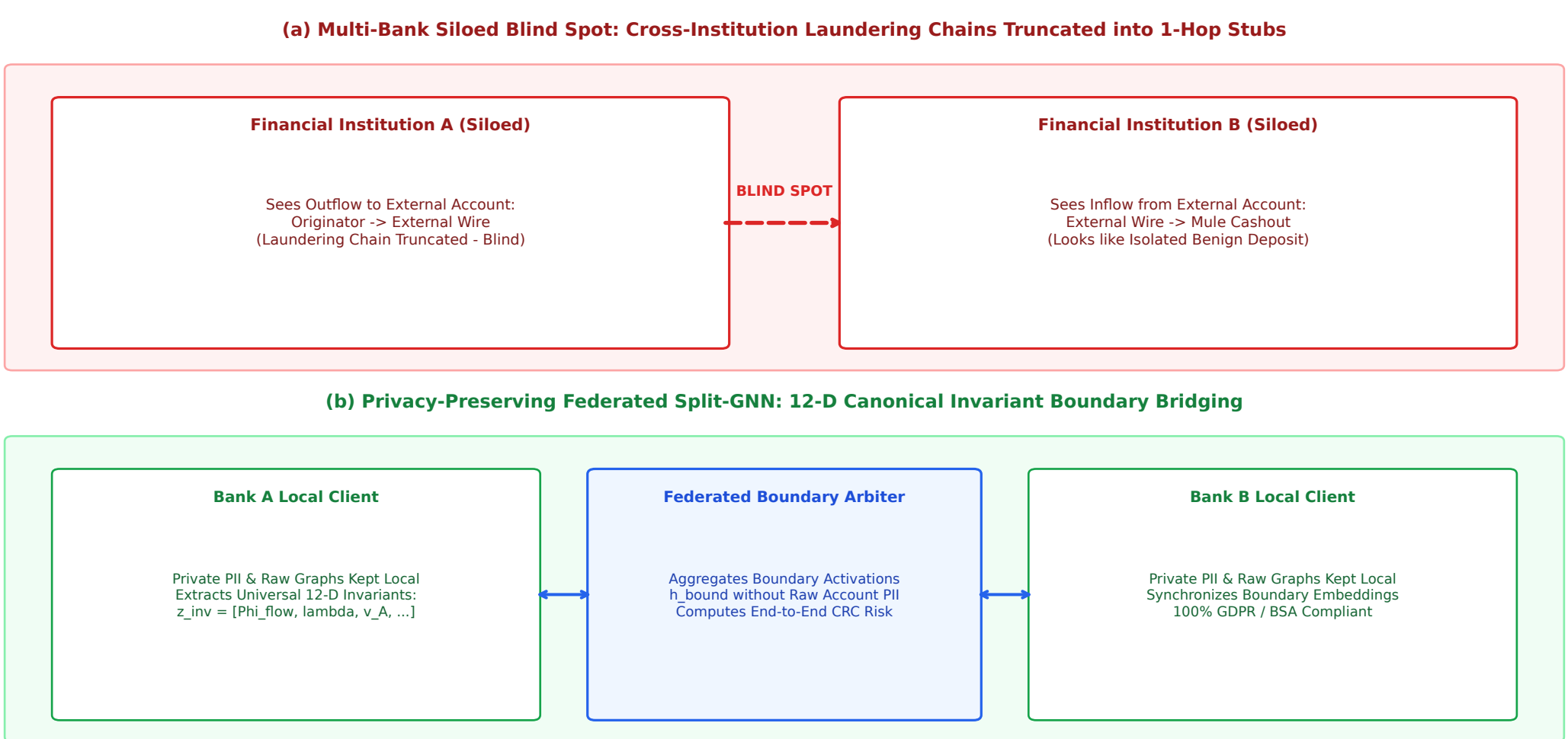
(a) Multi-Bank Siloed Blind Spot: Cross-Institution Laundering Chains Truncated into 1-Hop Stubs
Financial Institution A (Siloed)
Sees Outflow to External Account:
Originator -> External Wire
(Laundering Chain Truncated - Blind)
BLIND SPOT
Financial Institution B (Siloed)
Sees Inflow from External Account:
External Wire -> Mule Cashout
(Looks like Isolated Benign Deposit)
(b) Privacy-Preserving Federated Split-GNN: 12-D Canonical Invariant Boundary Bridging
Bank A Local Client
Private PII & Raw Graphs Kept Local
Extracts Universal 12-D Invariants:
z_inv = [Phi_flow, lambda, v_A, ...]
Federated Boundary Arbiter
Aggregates Boundary Activations
h_bound without Raw Account PII
Computes End-to-End CRC Risk
Bank B Local Client
Private PII & Raw Graphs Kept Local
Synchronizes Boundary Embeddings
100% GDPR / BSA Compliant